

PARUL UNIVERSITY

Faculty of Engineering & Technology
B.Tech – Semester III
Subject: Design of Data Structures (303105201)
Time: 2.5 Hours
Maximum Marks: 60

SECTION – A

(Total: 30 Marks)

Q1 (Compulsory) – 12 Marks

a) Answer ANY THREE of the following: ($3 \times 2 = 6$)

1. Define Algorithm.
2. What is Stack Overflow?
3. Define Dynamic Memory Allocation.
4. What is a Circular Queue?

b) Answer ANY THREE of the following: ($3 \times 2 = 6$)

1. Define Binary Tree.
 2. What is a Hash Function?
 3. Define Recursion.
 4. What is Load Factor?
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Q2 – 9 Marks

a) Explain stack operations with suitable diagram. (4)

b) Answer ANY ONE of the following: (5)

- Convert the infix expression $A + (B * C)$ into postfix form.
 - Evaluate the postfix expression: $5\ 6\ 2\ +\ *$
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Q3 – 9 Marks

- a) Explain Linear Search algorithm with example. (4)
- b) Answer ANY ONE of the following: (5)
- Explain Binary Search algorithm with suitable example.
 - Explain Interpolation Search.
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SECTION – B

(Total: 30 Marks)

Q4 (Compulsory) – 12 Marks

- a) Answer ANY THREE of the following: ($3 \times 2 = 6$)
1. Define Linked List.
 2. What is a Deque?
 3. Define AVL Tree.
 4. What is Breadth First Search (BFS)?
- b) Answer ANY THREE of the following: ($3 \times 2 = 6$)
1. Define Binary Search Tree (BST).
 2. What is Collision in hashing?
 3. Define Depth First Search (DFS).
 4. What is an Adjacency Matrix?
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Q5 – 9 Marks

- a) Explain insertion at beginning in a singly linked list with diagram. (4)
- b) Answer ANY ONE of the following: (5)
- Explain Doubly Linked List with diagram.
 - Explain Circular Linked List with suitable example.

Q6 – 9 Marks

- a) Explain Binary Tree traversals: Inorder, Preorder and Postorder. (4)
- b) Answer ANY ONE of the following: (5)
- Explain Binary Search Tree and its properties.
 - Explain Expression Tree with example.